

Exercise Session 13: Magnetic Fields in Matter 2

Question 1:

A very long solid cylindrical wire of radius a carries a steady current I , uniformly distributed across its cross-section. The wire is made of a linear magnetic material with susceptibility χ_m .

- Use Ampère's law in terms of $\mathbf{H}$ to determine the magnetic field intensity $\mathbf{H}(s)$ both inside ($s < a$) and outside ($s > a$) the wire.
- Hence determine the magnetic flux density $\mathbf{B}(s)$ everywhere. And find the magnetization $\mathbf{M}$ inside the wire.
- Determine the bound volume current density $\mathbf{J}_b$ and the bound surface current density $\mathbf{K}_b$.

Question 2:

A coaxial cable consists of two very long concentric cylindrical conductors of radii a and b , with $a < b$. The space between them is filled with a linear magnetic material of susceptibility χ_m . A current I flows along the inner conductor and returns along the outer conductor. Assume in each conductor that the current is uniformly distributed over its surface.

- Determine the magnetic field intensity $\mathbf{H}$ in the region $a < s < b$.
- Find the corresponding magnetic flux density $\mathbf{B}$ in the magnetic medium.
- Determine the magnetization $\mathbf{M}$ in the region between the conductors.

Question 3:

An ideal infinite solenoid has n turns per unit length and carries current I . Its interior is completely filled with a linear magnetic material of susceptibility χ_m .

- Determine the magnetic field intensity $\mathbf{H}$ inside the solenoid.
- Determine the magnetic flux density $\mathbf{B}$ inside the solenoid.
- Find the magnetization $\mathbf{M}$ of the material.
- Determine the bound surface current density on the magnetic material.
- State whether the bound current reinforces or opposes the free current for:
 - a paramagnetic material,
 - a diamagnetic material.

Question 4:

A sphere of linear magnetic material with susceptibility χ_m is placed in an otherwise uniform external magnetic field $\mathbf{B}_0 = B_0 \hat{z}$.

- a) State whether magnetization aligns with or opposes the applied field in each case: $\chi_m > 0$ and $\chi_m < 0$.
- b) Explain qualitatively why the magnetic field inside the sphere differs from the applied external field.
- c) Compare the physical behavior of a diamagnetic and a paramagnetic sphere placed in the same external field.
- d) Briefly discuss why this linear picture would fail for a ferromagnetic material.